

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Matrices

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2 . (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **ICSE 2025, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Answer: (c)

2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. **ICSE 2025, 4 marks**

SOLUTION

$$(a) A + C = \begin{bmatrix} 1-5 & 2+1 \\ 3+7 & 4-4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$$

$$(b) B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix} = \begin{bmatrix} -8+10 & 6+0 \\ -16+20 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$$

$$(c) 5B = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$$

$$(d) B(A + C) - 5B = \begin{bmatrix} 2-10 & 6-5 \\ 4-20 & 12-10 \end{bmatrix}$$

$$\text{Answer: (d)} \begin{bmatrix} -8 & 1 \\ -16 & 2 \end{bmatrix}$$

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